

Customer Support Ticket Analyser

Summary & Findings

Project Overview

The Customer Support Ticket Analyser is a Python-based project developed to manage, clean, and analyse customer support tickets. The system stores ticket information such as ticket number, customer name, issue description, and priority. It also allows new tickets to be added, validates ticket priorities, cleans text data, and extracts useful insights from customer issues.

Data Processing

The ticket descriptions were cleaned using Python string operations. Punctuation and unnecessary spaces were removed, text was converted to lowercase, and shorthand terms such as "ok" were replaced with "okay". This helped standardise the issue descriptions and made keyword analysis more consistent.

Key Findings

1. Ticket Volume

The analysis included the initial 10 tickets along with any additional tickets entered by the user. The total number of tickets was calculated automatically using the ticket list.

2. Priority Analysis

Tickets were classified into **High, Medium, and Low** priority categories. The number of tickets in each priority level was calculated using Python. This helps identify the distribution of support workload and the number of issues requiring higher attention.

3. Keyword Analysis

The system searched the cleaned issue descriptions for the keywords "**poor**", "**good**", "**slow**", and "**excellent**". The number of tickets containing each keyword was calculated using a reusable Python function.

This analysis helps identify common customer feedback patterns and service-related concerns.

4. Longest Issue Description

The program calculated the number of words in every cleaned issue description and identified the ticket with the highest word count. This provides an example of how text data can be measured and compared using Python.

5. Unique Words

A Python set was used to extract unique words from all ticket descriptions. The unique words were then sorted alphabetically. This provides a basic understanding of the vocabulary and topics appearing in customer support issues.

Conclusion

The project demonstrates how Python can be used to perform basic customer support ticket analysis. By combining dictionaries, lists, loops, functions, conditional statements, string operations, and sets, the system can clean ticket data and generate useful insights.

The analysis can help a support team understand **ticket priorities, common keywords, customer feedback patterns, and the nature of reported issues**. These techniques can also be extended in the future to include sentiment analysis, category-wise issue analysis, response-time analysis, and visual dashboards.